

Ballbot — Version 1 Design

A solar-assisted robot that finds tennis balls on a court, sweeps them into a hopper, and returns to its dock.

Prepared for Tim Parsa & Kevin Eisenbacher · July 17, 2026 · v1 concept — scoping draft

The whole game in version 1 is one loop done reliably: **see the balls** → **drive to them** → **lift them into a hopper** → **return to the dock**. Kevin's longer wishlist — serving gun, court squeegee, floor conditioner, debris blower, court-to-court travel — stays the north star but is deferred. This draft focuses on the two things you flagged: the **simplest possible pickup mechanism**, and **how it reaches balls against the fence and in the corners**. Schematics below.

1. The pickup mechanism — yes, like a bowling alley

Your instinct is exactly right, and it's the same trick a tennis ball machine already uses in reverse. A ball machine *launches* a ball by pinching it between **two counter-rotating rubber wheels**. Run those wheels the other direction and they become an **intake**: the ball gets grabbed at the bottom and thrown up a short chute into the hopper. It's the bowling-alley ball-return principle, shrunk to tennis-ball scale — and because ball machines already prove the hardware works at this size, it's low-risk.

How a ball gets from the ground into the hopper, front to back:

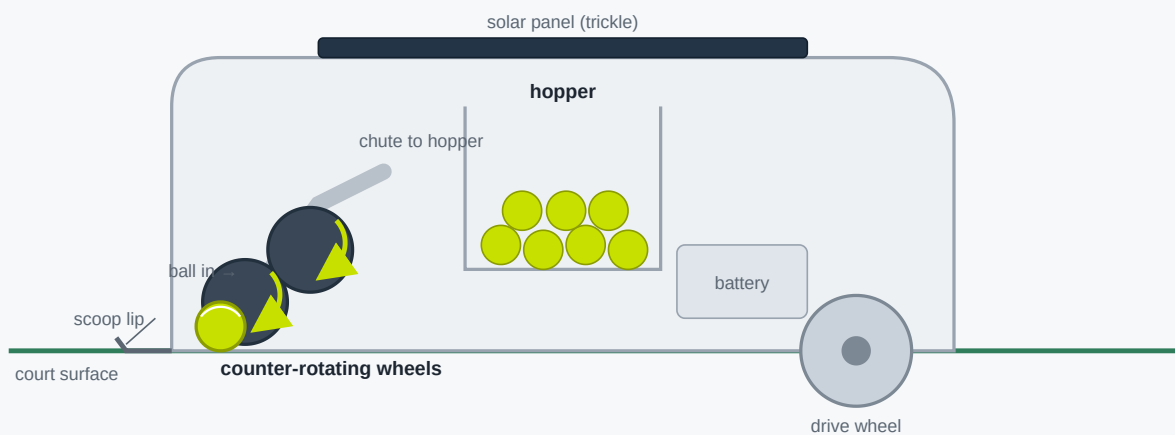
- **A low scoop lip** at the very front, riding ~2–3 mm off the court on a smooth skid. As the robot drives forward, a resting ball rolls up onto the lip instead of being pushed away.
- **A funnel throat** that narrows from a wide mouth down to the wheel gap, so the ball doesn't need to be perfectly centered — anything in the mouth gets steered into the nip.
- **Two counter-rotating wheels** (~70 mm foam-over-rubber), set a bit *closer* than a ball is wide (~50–55 mm gap). The ball (65 mm) is squeezed to ~52 mm, the wheels grip it, and their upward-moving faces fling it up the chute.
- **A short curved chute** carries the ball up and over into the hopper. One drive belt can spin both wheels off a single small motor.

Why this over a scoop or a plow: a passive scoop or ramp just shoves a ball ahead of it — the ball rolls away. Anything that reliably moves a ball *up and over* a lip needs to *grip* it, and

spinning wheels grip. Two wheels and one motor is about as simple as a reliable lift gets.

Simpler fallback: one brush roller

If we want to prototype even cheaper first, a single **spinning bristle roller** (like a powered push-sweeper or driving-range collector) can flick balls up the ramp with one motor and no precise gap to tune. It grips a fast, bouncy tennis ball a little less surely than the twin wheels, so the plan is to bench-test both cheaply and keep whichever proves more reliable. Lead design = twin wheels; brush = backup.



Side cross-section. Ball rolls onto the scoop lip, gets pinched by the two upward-spinning wheels (the ball-machine trick in reverse), and is flung up the chute into the hopper. One motor drives both wheels.

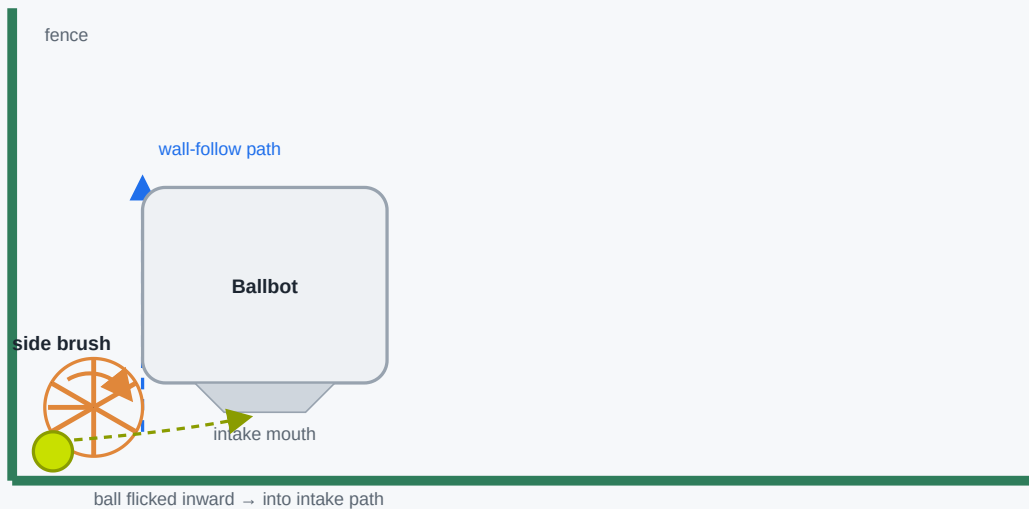
2. Reaching balls at the fence and in the corners

This is the real weakness of any front-mouth collector: it can't push its nose flush into a 90° corner, and a ball hugging the fence sits just outside the intake's path. The fix is borrowed from street sweepers and robot vacuums: **a rotating side-sweeper brush** mounted at the front corner of the robot, sticking out past the body edge.

- The robot runs a **"wall-follow" routine** along the fence when the overhead camera sees balls near it.
- The angled, vertical-axis **gutter brush** reaches into the corner and against the fence, and **flicks the ball inward** — away from the fence, into the robot's forward path where the twin-wheel intake grabs it.

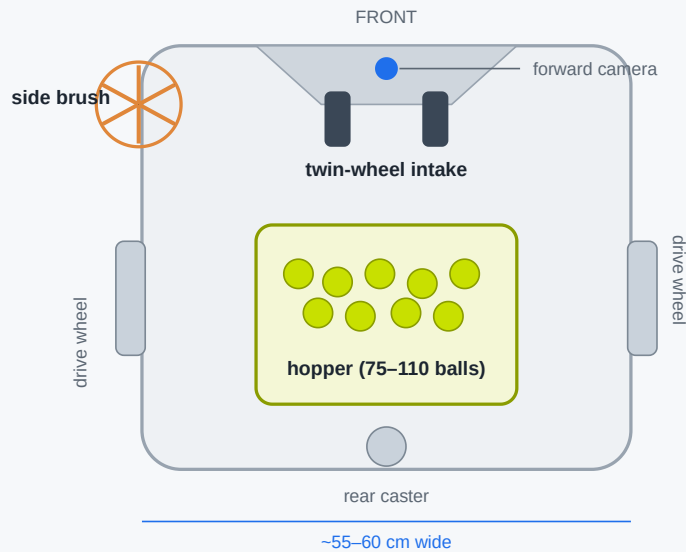
- It costs one extra small motor and is a thoroughly proven mechanism (every Roomba and street sweeper uses it).

For a true corner, the robot approaches diagonally so the brush can rake the ball out along one fence line first, then the intake collects it on the follow-through. Balls that are genuinely pinned in the apex are the hardest 1–2% — the brush handles nearly all of them; the rest we accept as a known limit for v1.



Top view of the corner routine. The robot follows the fence; the rotating side brush reaches past the body into the corner and flicks the ball inward into the intake's path.

3. Whole-robot layout (top view)

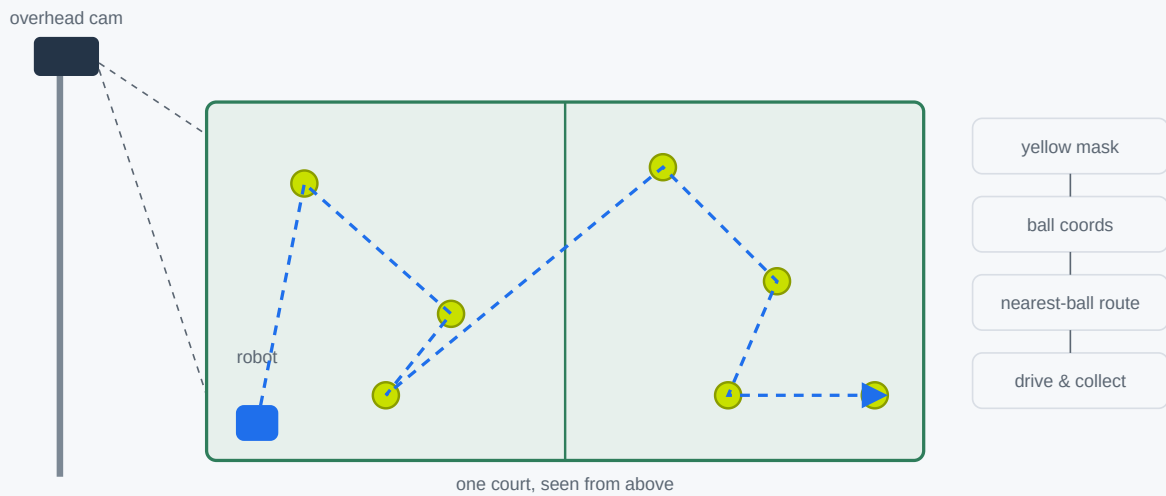


Top view. Twin-wheel intake at the front center, side brush on the front corner for fence/corner balls, hopper in the middle, differential drive with a rear caster, forward homing camera. Roughly 55–60 cm wide so it stays maneuverable on one court.

4. The camera + "find all the balls" program

Detection is the easy part, because optic yellow is the most distinctive color on a court — nothing else shares it. No heavy machine learning needed for v1; a straightforward color-vision pipeline does it:

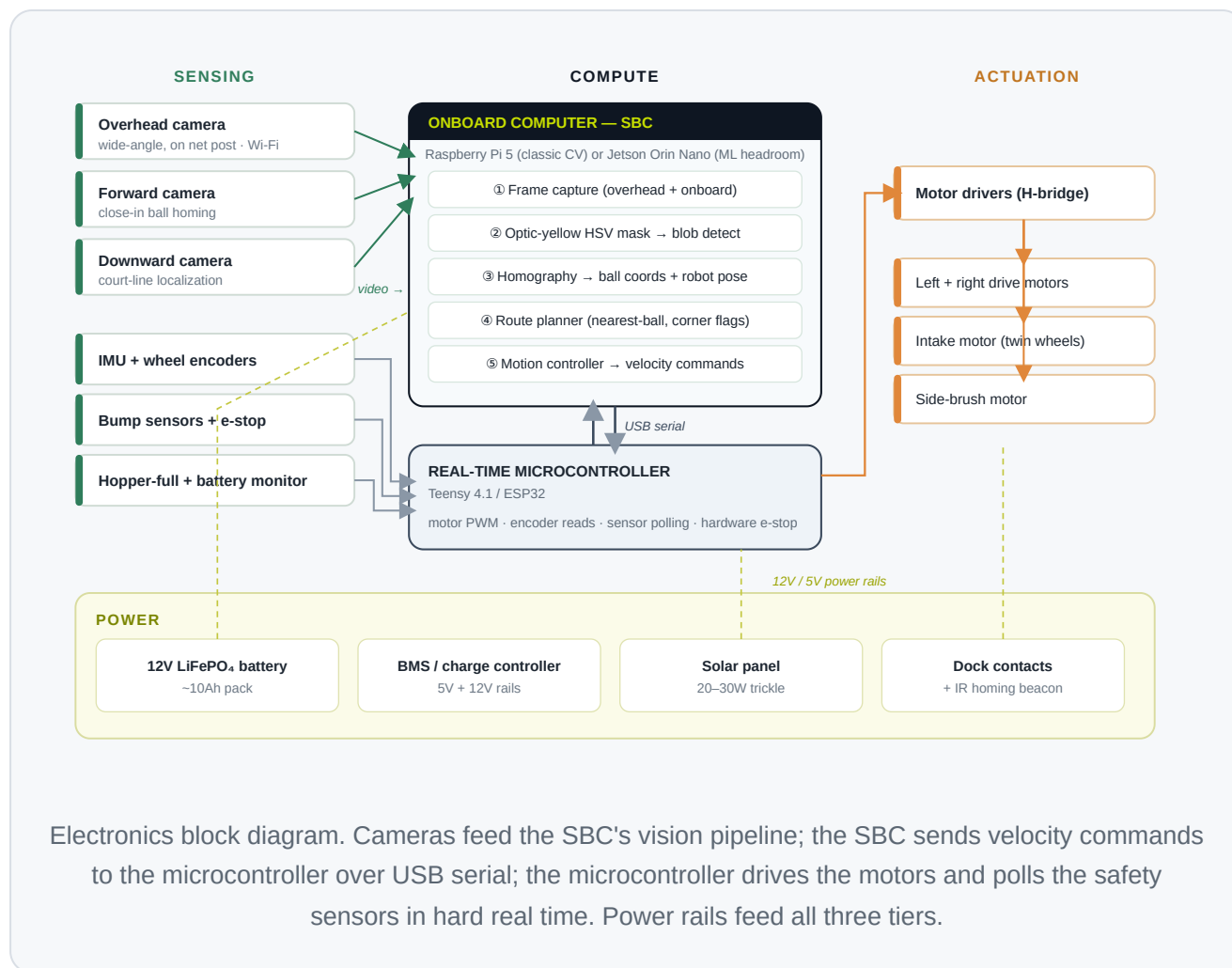
- **Overhead camera** on a net post or fence, looking down at the whole court, is the backbone — it sees every ball at once and tracks the robot too.
- **Color mask:** filter each frame for the optic-yellow band (HSV threshold) → clean blobs → a list of ball positions in court coordinates.
- **Route:** a simple nearest-ball-next plan (greedy, with corner balls flagged for the brush routine) orders the pickups to minimize driving.
- **Drive & collect:** robot drives to each ball; the **forward onboard camera** does the final centering into the intake mouth.
- **Fallback:** if a camera is blocked, the robot can still mow the whole rectangle in a back-and-forth pattern like a robot vacuum and sweep everything up.



The overhead camera maps every ball to court coordinates; a greedy nearest-ball route orders the pickups; the robot drives to each and the forward camera centers the final approach.

5. Electronics & compute architecture

Ballbot splits its computing the standard robotics way: a Linux single-board computer (the "brain") handles the heavy, non-real-time perception and planning, while a small microcontroller runs the hard-real-time motor-and-sensor loop. That split matters — it means a momentary hiccup in the vision pipeline can never stall a motor command or delay the e-stop. In v1 the overhead camera stays dumb: it just streams frames over Wi-Fi to the onboard computer, which does all the processing.



Electronics block diagram. Cameras feed the SBC's vision pipeline; the SBC sends velocity commands to the microcontroller over USB serial; the microcontroller drives the motors and polls the safety sensors in hard real time. Power rails feed all three tiers.

The compute split, concretely

- **Onboard computer (the brain):** a Linux SBC — a Raspberry Pi 5 is ample for the classic OpenCV color-vision pipeline; step up to a Jetson Orin Nano only if we later add a learned detector for tricky lighting or heavy occlusion. Runs frame capture, the optic-yellow mask, homography to court coordinates, the route planner, and the motion controller.
- **Real-time microcontroller:** a Teensy 4.1 or ESP32 handling motor PWM, quadrature encoder reads, sensor polling, and a hardware e-stop. Talks to the SBC over USB serial, taking velocity commands and returning odometry.
- **Overhead camera:** kept dumb in v1 — streams frames to the SBC over Wi-Fi. A later version could push detection onto a small edge board at the camera to cut bandwidth.
- **Vision stack:** capture → HSV optic-yellow mask → contour/blob detection → homography (pixel → court, calibrated once off the four court corners) → robot pose from an overhead marker fused with onboard odometry/IMU → greedy nearest-ball route with corner flags → velocity commands.

Bill of materials (v1 prototype, ballpark)

Subsystem	Component	Suggested part	~Cost
Compute	Onboard computer (brain)	Raspberry Pi 5 8GB or Jetson Orin Nano	\$80–250
Compute	Real-time microcontroller	Teensy 4.1 / ESP32	\$25
Vision	Overhead camera	Wide-angle IP cam / Pi HQ cam on Pi Zero 2 W	\$60
Vision	Forward camera	Pi Camera Module 3 / USB webcam	\$25
Vision	Downward camera	Small USB camera	\$15
Drive	Motor drivers	Dual H-bridge (Cytron MDD10 / VNH5019)	\$30
Drive	Drive motors ×2	12V gearmotors w/ encoders	\$40
Drive	Intake motor	12V brushed gearmotor	\$20
Drive	Side-brush motor	Small 12V gearmotor	\$12
Sensing	IMU	BNO085 / MPU-6050	\$15
Sensing	Bump, e-stop, hopper-full	Microswitches + IR break-beam / load cell	\$20
Power	Battery	12V LiFePO ₄ ~10Ah	\$60
Power	BMS / charge controller	Off-the-shelf BMS + DC-DC	\$20
Power	Solar panel	20–30W semi-flexible	\$40
Power	Dock	Charging contacts + IR homing beacon	\$25
Total	Ballpark, one prototype		≈ \$490–660

Rough prototype-quantity estimates; chassis, wheels, fasteners, and wiring extra. Not a production BOM.

6. Hopper — sizing to standard capacities

Rather than invent a size, we anchor to what teaching hoppers and carts already come in. Common commercial capacities: Gamma baskets ship in 50, 55, 75, 80, 90, 110, and 140; folding rolling teaching carts are typically 150. A standard lesson/drill basket is 50–75 balls, and a coach's cart is ~150.

Option	Capacity	Fits v1?
Small basket	50–55	Too many dump trips on a full court
Mid basket (recommended)	75–110	Best balance of size vs. dump frequency
Coach cart	140–150	Great capacity, but a taller/heavier robot

Recommendation: design the hopper cavity around a **standard 75–110 ball basket** — enough to clear a typical loose-ball session in one or two trips without making the robot tall and tippy. Keeping it to an off-the-shelf basket size also means the removable bin can literally be a stock hopper we drop in and lift out to empty.

7. Open decisions for Tim & Kevin

- **Intake bake-off:** build both the twin-wheel and single-brush intakes as cheap prototypes and pick the winner — agreed?
 - **Hopper target:** 75, 90, or 110 balls?
 - **Net:** treat each half-court as a separate job (my rec) vs. ducking under the 3-ft center strap.
 - **When it runs:** between sessions/drills (my rec) vs. during live play.
 - **Solar:** confirm it's a bonus trickle-charger, with the corner dock as primary power.
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8. Recommended v1 baseline

Fastest path to a demoable robot: **twin-wheel bowling-alley intake** with a funnel and scoop lip; a **front-corner side brush** plus wall-follow routine for fence and corner balls; an **overhead camera + simple yellow-blob detector** as the brain, with a forward camera for final homing; a **standard 75–110 ball hopper** as a drop-in bin; **half-court jobs, run**

between sessions, and **dock-primary power** with a token solar panel. Smallest machine that convincingly does the core loop — and the platform every later feature bolts onto.